



Lost but Not Only in the Middle

Positional Bias in Retrieval Augmented Generation

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Abstract. Large language models (LLMs) are known to exhibit positional bias, the tendency of models to perform differently based on where relevant information appears within the input context. Understanding this bias is important in a retrieval augmented generation (RAG) setting, as it impacts how retrieved passages are taken into account by the model. We systematically investigate positional bias in a RAG setting, by evaluating four LLMs using three different types of distractor documents, assessing their ability to extract relevant information from the input context. Our findings reveal significant positional bias depending on the type of context documents used and the total amount of documents in the context. Furthermore, the results show that positional bias in state-of-the-art LLMs is not limited to information located in the middle of the input context. By analyzing the models' attention, we identified patterns between the model's accuracy in responding to questions and the correct attribution of attention to relevant information in the context. Our code is available at <https://github.com/Janhutter/LBNOITM>.

1 Introduction

In this research, we will focus on retrieval augmented generation (RAG), an implementation where the performance of using a large context as input is particularly important. RAG is used to provide context to a language model with respect to a query that has been prompted and yields significant improvement over the base LLM [6]. The context consists of several documents retrieved from a collection, such as excerpts from Wikipedia. However, inputting the context documents into the prompt directly can result in reduced accuracy, as seen in [7]. Their research shows that placing one document which is relevant to the question, in the middle of a context with several 'distractor', or irrelevant, documents decreases accuracy on a question-answering task. It shows that the positioning of information inside the input is indicative of the performance of an LLM on knowledge-intensive tasks. This could be due to the mismatch between the task LLMs are trained on and the retrieval-augmented setup that is used during generation [10]. Parametric, or non-retrieval-augmented, LLMs are trained on datasets where the most informative tokens are the most recent ones, resulting in decreased accuracy when important information is placed in the middle of the input.

This paper investigates *how does positional bias impact the performance of retrieval-augmented large language models?* We conduct a systematic analysis of the occurrence of positional bias in the prototypical RAG setting of knowledge-intensive, open-domain question-answering tasks. Rather than a SotA study, we conduct a scientific experiment, where we manipulate the context input in a systematic way (our independent variables) and measure the effect on performance (our dependent variable). Specifically, we examine the influence of the position of the relevant document with respect to multiple distractor documents. We will refer to this influence of position of a relevant document within the context as positional bias. We systematically vary the position of the single relevant document among different contextual settings. These settings exist of three different types of distractor documents, and two context lengths of 5 and 10 documents. Specifically, these shorter contexts simulate a more reasonable scenario and allow for extended and complementary results of Liu et al. [7]. By exhausting all combinations of the positions of the relevant document, context types, and context lengths, this approach enables a controlled examination of positional bias on RAG performance. Finally, after examining the positional bias within each contextual setting, the inspection of the attention values of the used models will provide insights into the mechanisms underlying the manifestation of positional bias.

2 Related Research

The observation that modern transformers can naturally cope with many context documents for answer generation in open domain QA tasks was central to the development of RAG [2,3]. As Peysakhovich and Lerer [10] state: “Since the performance of the RAG system is bounded by the recall of the retriever and thus the number of documents returned, larger context windows can naturally improve the performance of these systems.” This highlights the importance of utilizing the context completely and increasing its size to improve performance in a RAG setting. Peysakhovich and Lerer [10] show that utilizing attention values of a prompted model to change the position of documents in the context can considerably improve performance, and reduce positional bias. By performing one or more steps of decoding and sorting the documents by the attention values that the model attributes to them, ensures that the documents are sorted by the inherent importance assigned by the model.

Our work is inspired by Liu et al. [7], who found that performance can degrade significantly when changing the position of relevant information in the long context input. They observed a *Lost in the Middle* effect with the best performance when relevant information occurs at the beginning or end of the input context. Similarly, Tang et al. [13] in *Found in the Middle* demonstrate positional bias in passage ranking tasks. Using five different LLMs, they show that their approach, *permutation self-consistency*, consistently improves document reranking and reduces the impact of positional bias. Documents that initially receive a lower score due to their position in the context are more accurately identified

and reranked with this method. Our paper contributes to this with an exhaustive analysis over different positions of the relevant information, context lengths, and distractor types, in the context of the task and models of a typical RAG setting [11].

Instead of changing only external aspects of the model, there is research on changing properties inside the architecture of LLMs to decrease positional bias. Yu et al. [16] conclude that “attention weights manifest position bias within transformers. Additionally, experiments demonstrate that, besides position embeddings, the causal mask also contributes to position bias.” By modifying attention values, the positional bias is reduced.

3 Method

To evaluate how the position of relevant information within a context affects the model’s performance in a question-answering setting, we systematically vary the position of the oracle document, the document that holds the information to answer the question, in the context of different types of distractor documents. Additionally, by varying the length of the context in which this oracle document is located, we assess the model’s ability to identify and interpret important information from varying lengths of text sections. Lastly, we will examine the internal attention values received during the generation of answers by the tested models to gain insights into positional bias. By analyzing the patterns in these attention values, we aim to uncover the underlying mechanisms that contribute to positional bias in LLMs.

In our approach to multi-context RAG, the inputs to the generative language model are as follows: (i) a question, (ii) a context consisting of n amount of documents where exactly one contains information relevant to the question, and $n-1$ distractor documents, and (iii) an instruction to help answer the question. The models should then, according to the instruction (iii), answer the question (i) while using relevant information from the context (ii). Building on these aspects, we construct a setup closely aligned with the one proposed by [7], to highlight positional bias in RAG. Specifically, we show this by controlling a single factor: the position of the document that contains the relevant information, which we will call the oracle document.

4 Experimental Setup

Datasets. To investigate the implications of positional bias in RAG, we make use of the KILT-NQ and KILT-HotpotQA datasets [9]. In our implementation, both datasets leverage the same collection for retrieval in the RAG setting. The collection consists of the publicly available KILT collection [9], where the documents are split into nonoverlapping chunks of 100 words, each prefixed with its title, which is similar to that found in Rau et al. [11]. The structure of the answers in the dataset differs from each other, increasing the generalizability. The NQ and HotpotQA datasets contain answers labels which are quite concise,

consisting of on average 3 and 2 words, respectively. NQ allows for more flexibility in answering, having on average two answer labels per question. Multiple answers are not allowed by HotpotQA. Another difference in answer labels of the datasets is the type of answer, where the NQ dataset has extractive answers and the HotpotQA dataset has short abstractive answers.

Context Types. By using different configurations of context type and position at which the oracle is injected, we can simulate a variety of scenarios. We have chosen three different context types, or distractor documents, in this research. First, *random distractor* consists of inserting the oracle document in a context comprised of uniformly sampled documents from the total collection, simulating a scenario where distractor documents are not relevant to the question. This represents the expected upper bound, as it assumes the easiest scenario for distinguishing relevant information from distractions. Secondly, *hard distractor* includes inserting the oracle document among relevant documents that do not contain the answer, simulating a realistic scenario where documents are closely related to the question but do not always provide the answer. These ‘relevant but incorrect’ documents are gathered using a BM25 retriever with a top-k of 1,000. To ensure that the retrieved documents did not contain the answer to the question for which they are retrieved, documents that contained the answer were removed from the ranking. Lastly, *relevant context* consists of inserting the oracle document among relevant documents retrieved by BM25, simulating a scenario where documents are related to the question and are able to contain the answer to the question. Resulting in a context where more than one document contains the answer. In related research like [7], context lengths of up to 30 documents are used. Their shortest context length is 10 documents, which we will use for comparison. However, increasing this context length given to different models from 5 to 10 or even 15 documents, no significant improvement in match and recall score was observed. Utilizing more documents in the context did show unwanted behavior: not all the models used in our research are able to tokenize the complete context because it exceeds the maximum length that these models support. For this reason, only context lengths of 5 and 10 documents are used in our experiments. Furthermore, increasing the context length can reduce effectiveness and efficiency, introducing the need for compression techniques [12].

Baselines. To consistently compare performance of the three ‘lost-in-the-middle’ test scenarios to an upper and lower bound, we make use of ‘closed book’ and ‘oracle’ settings. Closed book denotes a setting where the model is not provided with any additional information alongside the question; it relies solely on internally parameterized knowledge to generate an answer. This is the expected lower bound of our experiments. The oracle baseline represents a scenario where the LLM receives one or two documents per question, the oracle documents, that explicitly contain the answer, setting an expected upper bound for performance. These baselines help compare to scenarios when minimal positional bias occurs.

Prompting. After the selection of documents, an instruction is prepended to every question-context pair to inform the model about the method of generation.

For all the three aforementioned tests, this instruction is: “*Instruction: You are a helpful assistant. Your task is to extract relevant information from provided documents and to answer questions as short as possible.*” For the closed book baseline, the instruction was: “*Instruction: You are a helpful assistant. Answer the questions as short as possible.*” Using the combined context, four different decoder-only transformers are used to generate responses, which were then evaluated according to their similarity to the annotated answers, also called answer labels.

Metrics. Match Score is a variant of standard Exact Match [e.g., 8]. As models without fine-tuned decoders can generate verbose answers, the Match Score indicates whether the label is contained (as an exact match) in the generated answer. Because the Match Score is only accurate with short answer labels that consist of only a couple of words, recall is also used. Formally, in Match Score (Eq. 1) and Recall (Eq. 2) the amount of answer labels are denoted by M , the total amount of questions by N , the normalized answer labels split into a set of words by A_{ij} , and the normalized response of an LLM split into a set of words by R_i .

$$\text{Match Score} = \frac{1}{N} \sum_{i=1}^N \max_{j \in \{1, \dots, M\}} \left(\frac{|A_{ij} \cap R_i|}{|R_i|} \right) \quad (1)$$

$$\text{Recall} = \frac{1}{N} \sum_{i=1}^N \max_{j \in \{1, \dots, M\}} \left(\frac{|A_{ij} \cap R_i|}{|A_{ij}|} \right) \quad (2)$$

Utilizing these two metrics to assess model performance in a question-answering context will provide a robust approximation of their effectiveness and allow us to use them as indicators to evaluate their positional bias.

Models. To make sure the effects of positional bias are still present in modern decoder-based LLMs, we employ Mixtral-8x7B-Instruct [4], Llama-7B-chat-hf [14], and Solar-10.7-Instruct [5]. Figure 1 shows the performance of TinyLlama-1.1B-Chat [17] on the random distractor test, illustrating the extent to which positional bias can impact a model’s performance in a RAG setting. Seeing the performance of TinyLlama in Fig. 1, reveals significant positional bias in a smaller model, highlighting the need to explore this phenomenon in greater depth with the three larger models. The recall scores are omitted from Fig. 1, as the results demonstrated similar trends and results to those observed with the match score.

5 Experimental Results

Resilience to Changes in the Positioning of Essential Information. Figures 2 and 3 compare the per model performance on the three tasks. In the random distractor test utilizing a context length of 5 or 10 documents, the three larger models—Llama2, Solar, and Mixtral—show no significant positional bias. This shows that these models are able to discern important information in

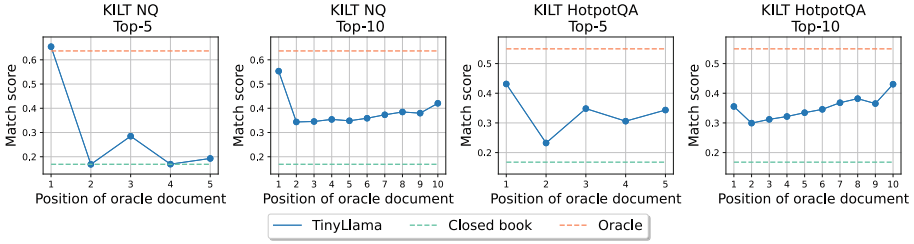


Fig. 1. The effect of changing the position of the oracle document in a context of random distractor documents, our ‘random distractor’ test scenario, when generating with TinyLlama on the NQ (left) and HotpotQA (right) datasets.

the input quite consistently. However, this consistency does not extend to the hard distractor and relevant context tasks, where the models show significant positional bias. We observe that these models show significantly decreased performance in both settings compared to the oracle setting. TinyLlama shows significant positional bias in all three test scenarios, showing large decrease in performance when the oracle document is not in the first position for all settings, except for the 10 document context on KILT-HotpotQA. In this setting, TinyLlama shows significantly increased performance when the oracle document is at the last position instead of the first.

A commonality between both context lengths is the increased performance of the Mixtral model on the random distractor task built from KILT-NQ compared to oracle performance. This same phenomenon was shown by TinyLlama in the 5 document random distractor task. These findings seem counterintuitive because adding irrelevant documents to the context seems to improve the performance of the Mixtral and TinyLlama models on the question-answering task.

The Figures show that all four models suffer from significant positional bias in both the hard distractor and relevant context tasks. The Llama2 and Solar models exhibit decreased performance when the relevant information appears at the end of the input (primacy bias), while Mixtral shows the opposite pattern, performing worse when the relevant information is at the beginning (recency bias). Specifically, Mixtral’s performance improves by 6.81% when the oracle document is at the last position instead of the first, in the hard distractor task with five documents on KILT-NQ. For the random distractor task under the same conditions, this fluctuation is more modest, at approximately 1.95%. For Llama2 the performance decreases by 12.45% in the 5 document hard distractor task on KILT-NQ, and decreases by only 1.16% in the random distractor task under the same conditions. The performance change in of the Solar model is less significant, decreasing 2.72% in the hard distractor task and 1.34% in the random distractor task.

Comparing the hard distractor and relevant context tests and both datasets side to side shows that the relevant context test has an average which is slightly higher for every model than that found on the hard distractor test. Similarly, we

see slightly less positional bias on the HotpotQA relevant context test than the HotpotQA hard distractor test. Furthermore, we see that the models show less positional bias on the HotpotQA than on the NQ dataset. In addition, using a larger context with 10 documents instead of 5 results in less significant changes in performance when changing the position of the oracle document in the context.

Figure 4 plots the results for the three larger models on the hard distractor task. The most notable result from this setting is the reversed ‘lost in the middle’ effect of Llama2 on the 5 document context constructed from HotpotQA. We see increased performance at positions 2, 3 and 4 compared to 1 and 5. This trend can also be found in the 10 document context, but having a smaller decrease in performance at the last positions as well. Similarly, Tynyllama showed increase performance in the middle (position 3) of the 5 document random distractor task when comparing to positions 1, 3, and 4.

Not Only ‘Lost in the Middle’. It is interesting to compare our findings for a typical RAG setting to the findings of [7]. There are some differences in the experimental setup. First, our results cover shorter context lengths at more frequent intervals. [7] use context lengths from 10 to up to 40 documents, evaluating every five positions. Our results cover every position, with up to 10 documents. When inspecting the types of distractor documents, [7] use documents which match our description of hard distractor, where documents were retrieved which were ‘most relevant to the query and do not contain any of the NaturalQuestions-annotated answers’.

For the typical RAG setting, our results show behavior which contradicts their title ‘Lost in the Middle’. The performance of the Mixtral model on the hard distractor and relevant context tests, when comparing to the other models, shows us that positional bias not only occurs in the middle of an LLM input context. Similarly, a positional bias of the Llama2 model on the HotpotQA hard distractor test was observed, which was the inverse of the ‘Lost in the Middle’ effect. Hence, the positional bias pattern depends on the used model, and not all LLMs exhibit the U-shaped performance over the position of relevant information in the input context.

6 Attention Analysis

Using the attention values of the models while generating, we can reason about what tokens in the input are important for a certain generated answer. Furthermore, the models’ attention values allow for a comparison between the model’s internal mechanisms and the external occurrence of positional bias. The attention values that are visualized are calculated during generation and are of the shape:

(attention steps, layers, batch size, number of heads, sequence length)

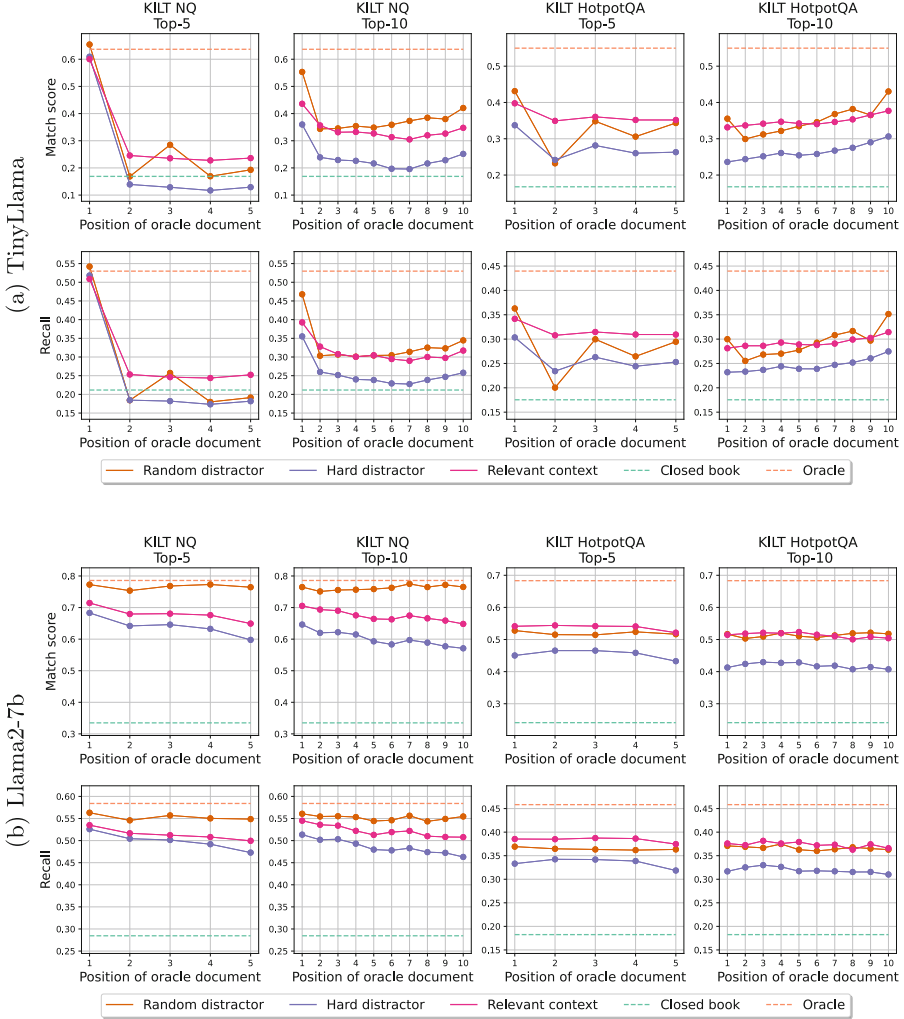


Fig. 2. Performance of TinyLlama and Llama2-7b models on the three different test scenarios constructed from KILT-NQ and KILT-HotpotQA.

We averaged over the attention steps for all the generated tokens, then took the attention for the middle and last attention layers. These values are then averaged over the number of attention heads. Consequently, using the amount of tokens for the instruction, documents, and question, their respective attention can be calculated. This allows us to see what the model calculated attention values are for every single document in the input context. The first layer is being left out in our results, as preliminary results showed that these layers only attend to the most recent document for every context type and model. This is similar to earlier findings: ‘The attention maps in the first two layers (layers 0

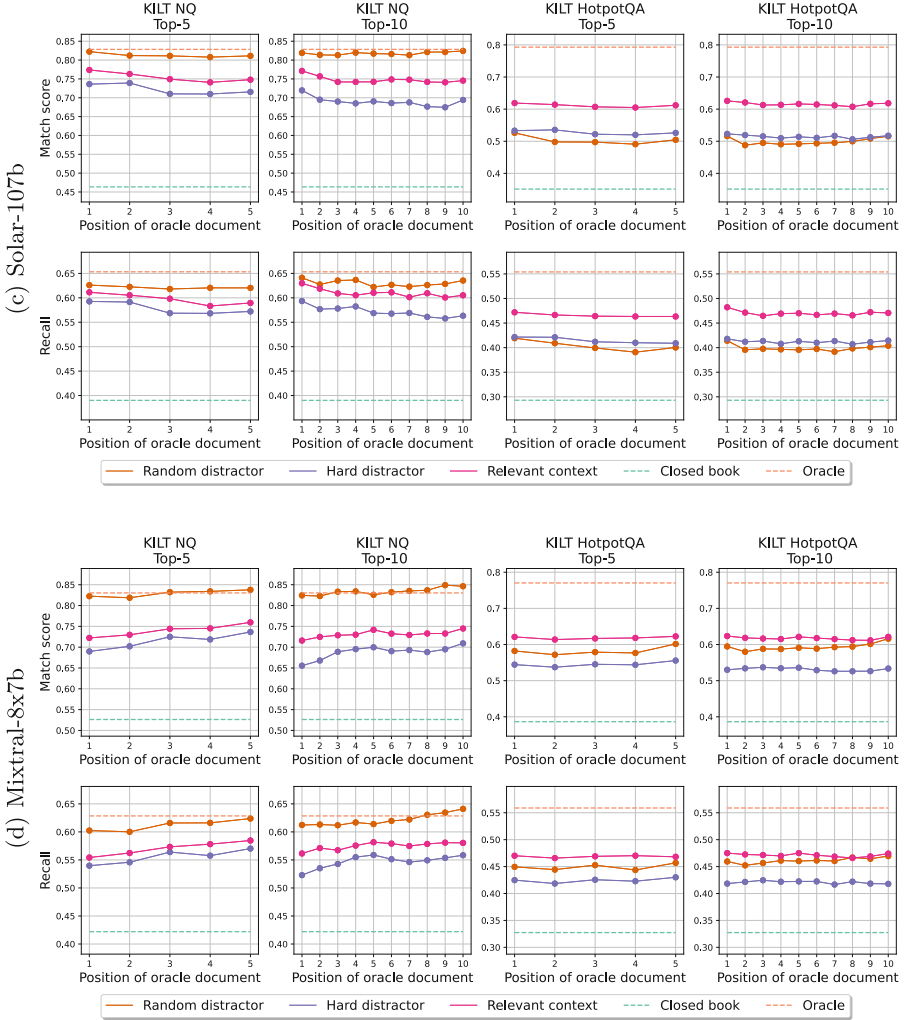


Fig. 3. Performance of Solar-107b and Mixtral-8x7b models on the three different test scenarios constructed from KILT-NQ and KILT-HotpotQA.

and 1) exhibit the “local” pattern, with recent tokens receiving more attention’ [15]. We mainly focus on the middle layer, which shows more interpretable results similar to the results by [10]. They found that ‘the differential attention to true vs. distractor documents happens mostly in the middle layers.’

This is similar to the visualizations made in [10], but where they only use the attention for the first generated token and averaged over the heads and layers. To distinguish the layers, all generated tokens are included to increase the amount of data points. Taking inspiration from their approach for mechanistic

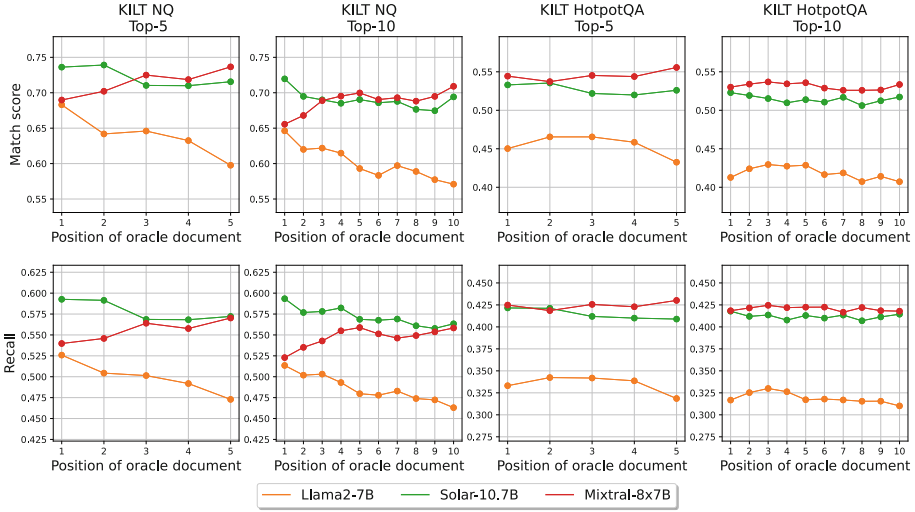


Fig. 4. The effect of changing the position of the oracle document in the context of hard distractor documents on the tested models for the NQ (left) and HotpotQA (right) datasets. We can observe the reversed ‘lost-in-the-middle’ effect of Llama2-7B on the KILT-HotpotQA dataset.

interpretability, the specific calculation listed above is utilized to look into TinyLlama and Llama2. Their attention values are discussed after generating 600 examples for every test scenario constructed from KILT-NQ and KILT-HotpotQA. We also differentiate our work by visualizing the model’s attention over the quality of the answer it is generating. This allows us to explore potential patterns between response quality and positional bias. Analyzing these patterns allows us to investigate how incorrect attention attribution might affect the correctness of the model’s response.

Using the generated examples and the models’ attention values, we will first look at the positions of documents which received the highest attention on average across the position of the oracle document in the context. Secondly, we will take 50 uniformly sampled questions and answers according to their correctness, where correctness is attributed by the match score. These visualizations will help look into the effect of the attribution of attention toward the correctness of the output of LLMs. The rationale for selecting 600 examples for analysis, instead of the whole dataset, was to reduce computational requirements. Furthermore, 600 samples would result in at least 50 correct and incorrect samples per test scenario for further analysis, where 50 samples would be enough to find regularities in the output attention values.

Attention Across the Answer Quality. On the random distractor test where TinyLlama showed strong positional bias, its attention values show a similarly large pattern between the quality of the answer and the fidelity of the attribution of attention towards the position of the oracle document, as seen in Fig. 5 (top).

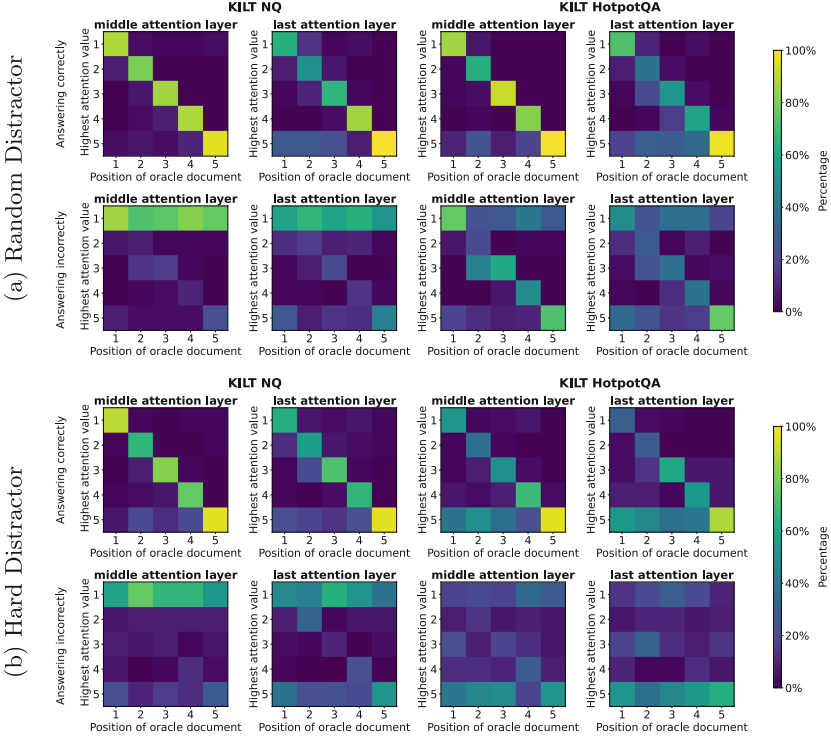


Fig. 5. The attention values of TinyLlama on the random distractor (top) and hard distractor (bottom) test. The variance in the position of the oracle document is plotted on the x-axis, and the position of the document with the most attention on average is plotted on the y-axis.

This pattern is stronger on KILT-NQ than KILT-HotpotQA. In the former, we see that the attention values of the middle layer follow the position of the oracle document closely when answering correctly. This pattern is displayed by the clear diagonal in the plot. When the answer to the question is incorrect, the model attributes most of its attention to the first document.

TinyLlama shows similar changes in attention values in the hard distractor task, as in the random distractor task as found in Fig. 5 (bottom). These results are very similar to those of the relevant context test, which have been omitted for brevity. However, TinyLlama shows higher attention values at the last position regardless of the position of the oracle document on the hard distractor task than the random distractor task. This finding contrasts with the findings of positional bias found in TinyLlama, where the model only showed primacy bias without significant changes over the different test scenarios.

Llama2 displays exceptionally consistent attention in the middle layer when changing the position of the oracle document in the random distractor test, found in Fig. 6 (top). This plot shows distinct colors for the five document positions,

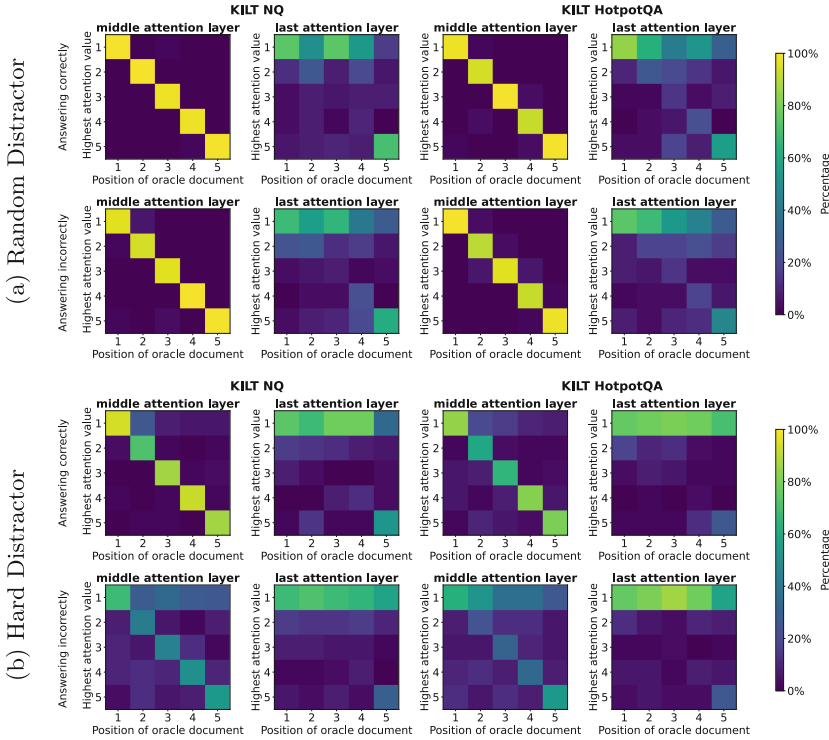


Fig. 6. The attention values of Llama2 on the random distractor (top) and hard distractor (bottom) test. The variance in the position of the oracle document is plotted on the x-axis, and the position of the document with the most attention on average is plotted on the y-axis.

as the oracle document receives the highest attention when varying its location. These results align with the minimal occurrence of positional in this setting as found in Fig. 2, meaning that in this setting and context length, there is a pattern between the amount of positional bias and the correct attribution of attention. Llama2 shows no pattern between the quality of the answer and the correctness of the attribution of attention. This would be explained by the insignificant amount of positional bias that this model shows on this test. Figure 6 (bottom) illustrates Llama2’s attention during the hard distractor task, which closely aligns with its attention in the relevant context task; the latter has been omitted for brevity. In both the hard distractor and relevant context tests, Llama2 demonstrates an attribution of attention that is less closely aligned with the position of the oracle document compared to the random distractor task.

7 Discussion and Conclusions

The goal of this research was to investigate positional bias in state-of-the-art LLMs, and expand upon the findings by [7] using RAG on two question answering datasets, both split into three identical tests.

We found that the smaller used model, TinyLlama, showed significant positional bias regardless of the type of distractor documents. The larger models showed positional bias to a lesser extent and could quite confidently distinguish the oracle document from the random distractors. The models showed quite significant positional bias on the hard distractor and relevant context tests. Both Llama2-7b-chat-hf and Mixtral-8x7b-instruct showed positional bias in the input context locations, unlike the findings of [7]. Rather than only *Lost in the Middle*, we observed varying patterns better described as *Lost but Not Only in the Middle*. More generally, understanding the exact positional bias of particular models in a RAG setting is important when designing a RAG framework for real-life applications.

Our findings are supported by the analysis of the attention values in TinyLlama and Llama2, revealing a significant pattern between the correctness in the placement of attention and the quality of the answer. This finding would be explained by the models incorrectly identifying relevant information in the context using its attention mechanism, which then causes decreased performance, or positional bias. However, we need to be careful with describing this as the actual cause of positional bias, with other research doubting the truthfulness of using attention as an explanation [1].

Concluding, our results both quantify the impact and importance of positional bias in RAG, but also show that further research on the phenomenon of positional bias in LLMs is needed. In future research, we are extending the experiments to further models and other RAG settings (including the TREC RAG track datasets). We are also extending the analysis of attention and understanding of how this impacts the models internally, as our controlled experiments relating attention to performance can also help understand the types of explanations we can derive from such attention analysis [1, e.g.,].

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